

Paying for electricity

Each home contains an **electricity meter** that records the transfer of energy from the National Grid to the house. All electricity supplied to the house passes through the meter.

We know that $P = \frac{W}{t}$ and therefore $W = Pt$. We can use this to calculate the energy transferred by an electrical device.

The kWh

It is inconvenient to use the joule when measuring the energy transferred to our homes. It takes at least 100 kJ to boil just one cup of water. Electricity bills therefore use a derived unit, the **kilowatt-hour** (kWh), defined as the energy transferred by a device with a power of 1 kW operating for a time of 1 hour.

$$1 \text{ kWh} = 3.6 \text{ MJ}$$

The cost of each kW of electrical energy (sometimes simply referred to as a 'unit') varies between operating companies. It depends on the particular tariff from the electricity company, and even in some cases on the time of day the energy is transferred. Typical costs are around 6-15p per kWh.

Summary Questions

- $W = Pt = (2 \times 60 \times 60\text{s}) \times 60\text{W} = 432,000\text{J} = 432 \text{ kJ}$.
- $0035387 - 0034512 = 875 \text{ units}$. $875 \times \text{£}0.12 = \text{£}105$
- $1 \text{ kWh} = 1000 \text{ Wh} = 1000(\text{J/s})\text{h} = 1000 \times \frac{3600\text{s} \times \text{J}}{\text{s}} = 3600000\text{J}$
- $9000\text{W} = 9000 \text{ J/s}$. $9000 \text{ J/s} \times 3600 = 32400000\text{J}$
 - $9000\text{W} = 9 \text{ kW}$. $9 \text{ kW} \times \frac{15}{60} = 2.25 \text{ kWh}$.
- $5.0 \text{ weeks} = 5.0 \times 7 \times 24\text{h} = 840\text{h}$.
 $\frac{60}{1000} \text{ kW} \times 840\text{h} \times \text{£}0.112 = \text{£}5.64$
- Over one year with filament lamps:

$$\text{total wattage} = 18 \times 100\text{W} = 1.8 \text{ kW}.$$

$$\begin{aligned} \text{energy transferred in one year} &= 365.25 \times 2 \times 1.8 \\ &= 1314.9 \text{ kWh} \end{aligned}$$

Over one year with LEDs:

$$\text{total wattage} = 18 \times 15\text{W} = 0.27 \text{ kW}.$$

$$\begin{aligned} \text{energy transferred in one year} &= 365.25 \times 2 \times 0.27 \\ &= 197.235 \text{ kWh} \end{aligned}$$

Therefore the difference in energy is 1117.665 kWh, or 4 GJ.